

NINETEEN VECTORS AND THEIR PHASE-RETRIEVAL INJECTIVITY IN $\mathbb{C}^6$

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ABSTRACT. We exhibit nineteen explicit vectors in $\mathbb{C}^6$, with Gaussian-integer entries, that do phase retrieval: the intensities $|\langle a_j, x \rangle|^2$, $j = 1, \dots, 19$, determine every $x \in \mathbb{C}^6$ up to a unimodular scalar. This gives $m_{\mathbb{C}}(6) \leq 19 = 4d - 5$, one below the generic count $4d - 4$; combined with the lower bound $m_{\mathbb{C}}(6) \geq 18$ of Wang and Xu, the minimal measurement number in dimension six is 18 or 19. The claim is verified by an exact rational computation, archived and rerunnable.

1. STATEMENT

Fix measurement vectors $a_1, \dots, a_m \in \mathbb{C}^d$ and let

$$\Phi_A(x) = (|\langle a_1, x \rangle|^2, \dots, |\langle a_m, x \rangle|^2), \quad \langle a, x \rangle = \sum_k \bar{a}_k x_k.$$

The family *does phase retrieval* if $\Phi_A(x) = \Phi_A(y)$ forces $y = \omega x$ with $|\omega| = 1$; $m_{\mathbb{C}}(d)$ denotes the minimal m for which such a family exists in $\mathbb{C}^d$. Generic families of $4d - 4$ vectors do phase retrieval [2], and Vinzant [4] exhibited $11 = 4 \cdot 4 - 5$ vectors doing phase retrieval in $\mathbb{C}^4$. In dimension six, Wang and Xu proved $m_{\mathbb{C}}(6) \geq 18$ [5, Theorem 5.2], while the best upper bound known has been the generic $4d - 4 = 20$; whether $19 = 4d - 5$ vectors can do phase retrieval in $\mathbb{C}^6$ has been open. We answer affirmatively.

Theorem 1. *The nineteen row vectors of Table 1 do phase retrieval in $\mathbb{C}^6$. Consequently $18 \leq m_{\mathbb{C}}(6) \leq 19$.*

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A6 = [
[1, -5, 25, -125, 625, -3125],
[1, -4, 16, -64, 256, -1024],
[1, -3, 9, -27, 81, -243],
[1, -2, 4, -8, 16, -32],
[1, -1, 1, -1, 1, -1],
[1, 0, 0, 0, 0, 0],
[1, 1, 1, 1, 1, 1],
[1, 2, 4, 8, 16, 32],
[1, 3, 9, 27, 81, 243],
[1, 4, 16, 64, 256, 1024],
[1, 5, 25, 125, 625, 3125],
[9765625, 13671875+46875000j, -205859375+131250000j,
-918203125-804375000j, 2575515625-5533500000j, 30166521875+4615575000j],
[9765625, 10937500+37500000j, -131750000+84000000j,
-470120000-411840000j, 1054931200-2266521600j, 9884965888+1512431616j],
[9765625, 8203125+28125000j, -74109375+47250000j,
-198331875-173745000j, 333786825-717141600j, 2345748741+358907112j],
[9765625, 5468750+18750000j, -32937500+21000000j,
-58765000-51480000j, 65933200-141657600j, 308905184+47263488j],
[9765625, 2734375+9375000j, -8234375+5250000j,
-7345625-6435000j, 4120825-8853600j, 9653287+1476984j],
[9765625, -2734375-9375000j, -8234375+5250000j,
7345625+6435000j, 4120825-8853600j, -9653287-1476984j],
[9765625, -8203125-28125000j, -74109375+47250000j,
198331875+173745000j, 333786825-717141600j, -2345748741-358907112j],
[9765625, -10937500-37500000j, -131750000+84000000j,
470120000+411840000j, 1054931200-2266521600j, -9884965888-1512431616j],
]
```

TABLE 1. The nineteen measurement vectors $a_1, \dots, a_{19}$ for $\mathbb{C}^6$ (rows; $j = \sqrt{-1}$).

Date: August 2026.

2020 *Mathematics Subject Classification.* 42C15, 94A12.

2. VERIFICATION

The verification is a finite, exact computation; this section records the elementary reduction that produces the computed system, and the archived facts about that system.

Reduction to polynomials. Every row of Table 1 is a scalar multiple of a vector $a(w) := (1, \bar{w}, \bar{w}^2, \dots, \bar{w}^5)$: rows 1–11 are $a(r)$ with $r = -5, \dots, 5$, and rows 12–19 are $25^5 a(\zeta s)$ with

$$\zeta = \frac{-7+24i}{25}, \quad s \in \{-5, -4, -3, -2, -1, 1, 3, 4\},$$

as one checks by expanding the powers. Scaling a measurement vector by a nonzero constant scales the corresponding intensity by a positive constant, so it suffices to verify phase retrieval for the unscaled family $\{a(r)\} \cup \{a(\zeta s)\}$. For $x \in \mathbb{C}^6$ let $f_x(z) = \sum_{k=0}^5 x_{k+1} z^k$, so that $\langle a(w), x \rangle = f_x(w)$; the map $x \mapsto f_x$ is a linear bijection onto the polynomials of degree ≤ 5 . For $f(z) = \sum c_k z^k$ write $f^\#(z) = \sum \bar{c}_k z^k$; then $f^\#(r) = \overline{f(r)}$ for real r , $|f(z)|^2 = f(z)f^\#(\bar{z})$ in general, and $(f^\#)^\# = f$.

For real polynomials U, V put

$$\mathcal{W}(U, V)(t) = \frac{1}{2i} \left(U(\zeta t) V(\bar{\zeta} t) - U(\bar{\zeta} t) V(\zeta t) \right).$$

Its values at real t are real, so its coefficients are real; if $\deg U, \deg V \leq k$, the coefficient of t^m collects the terms $u_i v_j - u_j v_i$ with $i < j$, $i + j = m$, so $\deg \mathcal{W}(U, V) \leq 2k - 1$ and $t \mid \mathcal{W}(U, V)$; the coefficient of t^1 is $\sin \theta (u_1 v_0 - u_0 v_1)$ with $\theta = \arg \zeta$, $\sin \theta = \frac{24}{25}$. A direct expansion with $u = U + iV$, $u^\# = U - iV$ gives, for real s ,

$$(1) \quad |u(\zeta s)|^2 - |u^\#(\zeta s)|^2 = u(\zeta s) u^\#(\bar{\zeta} s) - u(\bar{\zeta} s) u^\#(\zeta s) = 4 \mathcal{W}(U, V)(s).$$

Two elementary facts. *First:* if $f, g \neq 0$ have degree ≤ 5 and $|f| = |g|$ at the eleven points $r = -5, \dots, 5$, then $f f^\# - g g^\#$, of degree ≤ 10 , vanishes at eleven points, so $f f^\# = g g^\#$; writing $h = \gcd(f, g)$, $f = hu$, $g = hv$ with $\gcd(u, v) = 1$ and cancelling $hh^\#$ gives $uu^\# = vv^\#$, whence v divides $u^\#$ and u divides $v^\#$, so by degrees $v = \lambda u^\#$ with $|\lambda| = 1$: every such pair has the form

$$(2) \quad f = hu, \quad g = \omega h u^\#, \quad |\omega| = 1, \quad \deg h + \deg u \leq 5.$$

Second: if real U, V satisfy $\mathcal{W}(U, V) \equiv 0$, they are linearly dependent. Indeed ζ^2 is not a root of unity — otherwise ζ would be one, making $\zeta + \zeta^{-1} = 2 \operatorname{Re} \zeta = -\frac{14}{25}$ a rational algebraic integer, hence an integer — and for coprime U, V the identity $U(\zeta t) V(\bar{\zeta} t) = U(\bar{\zeta} t) V(\zeta t)$ forces $U(\zeta t) \mid U(\bar{\zeta} t)$ (a common root τ of $U(\zeta t), V(\zeta t)$ would make $\zeta \tau$ a common root of U, V), so $U(\bar{\zeta} t) = \lambda U(\zeta t)$ and the root multiset of U is invariant under multiplication by ζ^2 , which has infinite order: $U = u t^p$, symmetrically $V = v t^q$, $\min(p, q) = 0$ by coprimality, and substituting back forces $\zeta^{2(p-q)} = 1$, so $p = q = 0$ and both are constants.

Proposition 2. *The unscaled family fails phase retrieval if and only if there exist real linearly independent U, V of degree ≤ 5 and a real $\gamma \neq 0$ with*

$$(3) \quad \mathcal{W}(U, V)(t) = \gamma t(t+5)(t+4)(t+3)(t+2)(t+1)(t-1)(t-3)(t-4).$$

Moreover every such pair satisfies $u_1 v_0 - u_0 v_1 \neq 0$.

Proof. ($\Leftarrow$) Set $u = U + iV$ and let x, y be the coefficient vectors of $u, u^\#$. Then $|u(r)| = |u^\#(r)|$ for all real r , and at the eight rotated points (1) and (3) give $|u(\zeta s)| = |u^\#(\zeta s)|$; so all nineteen measurements agree. If $u^\# = \omega u$ with $|\omega| = 1$, then $(1 - \omega)U = i(1 + \omega)V$ would make U, V dependent; hence y is not a unimodular multiple of x , and phase retrieval fails.

($\Rightarrow$) Let $f = f_x$, $g = f_y$ witness a failure. If $f = 0$, then g vanishes at eleven > 5 real points, so $g = 0 = f$, a contradiction; so $f, g \neq 0$ and (2) applies, with $u \nparallel u^\#$ (else g is a unimodular multiple of f), i.e. U, V independent for $u = U + iV$. Let $k = \deg u$, so $\deg h \leq 5 - k$. At each rotated point with $h(\zeta s) \neq 0$, dividing $|f(\zeta s)| = |g(\zeta s)|$ by $|h(\zeta s)|$ and using (1) gives $\mathcal{W}(U, V)(s) = 0$; since h has at most $5 - k$ roots, at least $8 - (5 - k) = k + 3$ of the eight values s are roots of $\mathcal{W} := \mathcal{W}(U, V)$. Write $\mathcal{W} = t \widetilde{\mathcal{W}}$, $\deg \widetilde{\mathcal{W}} \leq 2k - 2$. If $k \leq 4$, then $k + 3 > 2k - 2$: $\widetilde{\mathcal{W}}$ has more distinct nonzero roots than its degree, so $\mathcal{W} \equiv 0$ and the second fact above makes U, V dependent — a contradiction. Hence $k = 5$, h is constant, all eight

equations survive, and $\widetilde{\mathcal{W}}$, of degree ≤ 8 , has the eight listed roots: $\widetilde{\mathcal{W}} = \gamma \prod (t - s)$ with $\gamma \neq 0$ (else $\mathcal{W} \equiv 0$ again), which is (3). Comparing coefficients of t^1 in (3) gives $\frac{24}{25}(u_1 v_0 - u_0 v_1) = \gamma \prod_s (-s) \neq 0$. $\square$

The computation. Equation (3) is linear in the Plücker coordinates $q_{ij} = u_j v_i - u_i v_j$ of the plane spanned by U, V . In the chart $q_{01} = u_1 v_0 - u_0 v_1 = 1$, legitimate by the last clause of Proposition 2, the coefficient equations of (3) reduce to five quadratic equations in five unknowns with rational coefficients — written out in full, with their derivation, in Appendix A. The archived computation [3] certifies, in exact rational arithmetic with no floating point:

- the system is zero-dimensional, and its eliminant with respect to one coordinate is a squarefree integer polynomial of degree fourteen (matching the Bézout count for a proper linear section of the Plücker-embedded $\text{Gr}(2, 6)$, whose degree is $C_4 = 14$);
- the Sturm sequence of the eliminant has seven sign variations at $-\infty$ and seven at $+\infty$: the eliminant has *no real roots*. A real solution of the system would make its eliminated coordinate a real root; hence no real solution exists, no pair (U, V, γ) as in Proposition 2 exists, and Theorem 1 follows.

Protocol: the Gröbner-basis computation was performed with `msolve` [1] over $\mathbb{Q}$; the real-root count is independently re-derived from the archived eliminant by a Sturm-sequence script in integer arithmetic; the polynomial system was generated by two independently written emitters and compared as canonically normalized sets before solving; as a sanity check, the 19×36 real matrix of the lifted measurements $\bar{a}_j a_j^\top$ has rank 19 over $\mathbb{Q}$; and the same pipeline, run on configurations engineered to fail, reports the expected nonzero real-solution counts. The `msolve` input, the certificate data (eliminant, Sturm sequence, sign counts, SHA-256 hashes), and a standalone exact verifier are archived at [3] (directory `certificate/`); the verification reruns in minutes on a laptop.

Acknowledgment. The author received assistance from an AI assistant in the search for candidate frames and in the preparation of this manuscript.

APPENDIX A. THE CERTIFIED SYSTEM

This appendix specifies the certified system exactly as archived (`certificate/certificate.json`); the verifier (`certificate/verify_exact.py`) rebuilds it from the description below and rechecks every claim.

Write $U = \sum_{i=0}^5 u_i t^i$, $V = \sum_{i=0}^5 v_i t^i$ and $q_{ij} = u_j v_i - u_i v_j$. Since $(\zeta^i \bar{\zeta}^j - \bar{\zeta}^i \zeta^j)/2i = \sin((i-j)\theta)$ with $\theta = \arg \zeta$, the coefficients of $\mathcal{W}$ are

$$[t^m] \mathcal{W}(U, V) = \sum_{i < j, i+j=m} s_{j-i} q_{ij}, \quad s_k := \sin k\theta = \text{Im } \zeta^k,$$

and exactly

$$s_1 = \frac{24}{25}, \quad s_2 = -\frac{336}{625}, \quad s_3 = -\frac{10296}{15625}, \quad s_4 = \frac{354144}{390625}, \quad s_5 = \frac{1476984}{9765625}.$$

Let

$$g(t) = \prod_s (t - s) = t^8 + 7t^7 - 16t^6 - 182t^5 - 91t^4 + 1183t^3 + 1546t^2 - 1008t - 1440,$$

the product over the eight listed s , with coefficients $g_0, \dots, g_8$ in ascending order.

Chart. A witness (U, V, γ) of Proposition 2 has $q_{01} \neq 0$, and replacing (U, V) by a basis of the same plane with change-of-basis determinant q_{01}^{-1} scales every q_{ij} , hence $\mathcal{W}$ and γ , by q_{01}^{-1} ; so we may take $q_{01} = 1$. Equation (3) then says

$$\sum_{i < j, i+j=m} s_{j-i} q_{ij} = \gamma g_{m-1}, \quad m = 1, \dots, 9.$$

The $m = 1$ equation gives $\gamma = s_1/g_0$ (so $\gamma \neq 0$ is automatic). In this chart the three-term Plücker relation on the indices $(0, 1, i, j)$ gives $q_{ij} = q_{0i}q_{1j} - q_{0j}q_{1i}$ for $2 \leq i < j \leq 5$, so the eight coordinates q_{0k}, q_{1k} ($k = 2, \dots, 5$) determine the plane. The equations $m = 2, 3, 4$ are linear and solve to

$$q_{02} = \frac{s_1 g_1}{s_2 g_0} = -\frac{5}{4}, \quad q_{12} = \frac{g_2}{g_0} - \frac{s_3}{s_1} q_{03}, \quad q_{13} = \frac{s_1 g_3/g_0 - s_4 q_{04}}{s_2}.$$

Five unknowns remain; following the archived naming, put

$$\beta_4 = q_{03}, \quad \beta_5 = q_{04}, \quad \beta_6 = q_{05}, \quad \alpha_5 = -q_{14}, \quad \alpha_6 = -q_{15}$$

(and $\beta_3 = q_{02}$, $\alpha_3 = -q_{12}$, $\alpha_4 = -q_{13}$ for the quantities already determined above). Substituting into the equations $m = 5, \dots, 9$ and clearing denominators to primitive integer form yields the archived system, verbatim:

$$\begin{aligned} E_1 : & 617760000 \alpha_5 - 617760000 \beta_4^2 + 966250000 \beta_4 - 1897200000 \beta_5 \\ & + 141790464 \beta_6 - 1707265625 = 0, \\ E_2 : & -225000 \alpha_5 - 303552 \alpha_6 + 123552 \beta_4 \beta_5 - 193250 \beta_5 - 40625 = 0, \\ E_3 : & -225000000 \alpha_5 \beta_4 - 193050000 \alpha_6 + 106007616 \beta_4 \beta_6 - 379440000 \beta_5^2 \\ & - 330078125 \beta_5 - 165808500 \beta_6 - 2500000 = 0, \\ E_4 : & 360000 \alpha_6 \beta_4 + 607104 \beta_5 \beta_6 + 528125 \beta_6 + 3125 = 0, \\ E_5 : & 1440 \alpha_5 \beta_6 - 1440 \alpha_6 \beta_5 + 1 = 0. \end{aligned}$$

A real witness of Proposition 2 produces, through the (real) normalization above, a real solution of $E_1 = \dots = E_5 = 0$.

The eliminant. Let $I \subset \mathbb{Q}[\beta_4, \beta_5, \alpha_5, \alpha_6, \beta_6]$ be the ideal generated by $E_1, \dots, E_5$. The archived eliminant $E(\beta_5)$ is the primitive squarefree integer polynomial of degree 14 generating $I \cap \mathbb{Q}[\beta_5]$, found via a grevlex Gröbner basis followed by FGLM conversion to lexicographic order — but no property of that computation is needed beyond the certified fact $E \in I$, which the verifier checks by exact reduction of E to zero against the archived basis. Membership alone yields the logic of the certificate: at any solution of the system, real or complex, the β_5 -coordinate is a root of E ; the Sturm sequence of E has seven sign variations at $-\infty$ and seven at $+\infty$, so E has no real roots; hence the system has no real solutions. (In particular, neither zero-dimensionality nor completeness of the elimination is used in this direction; both hold and are certified, but only as cross-checks.)

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